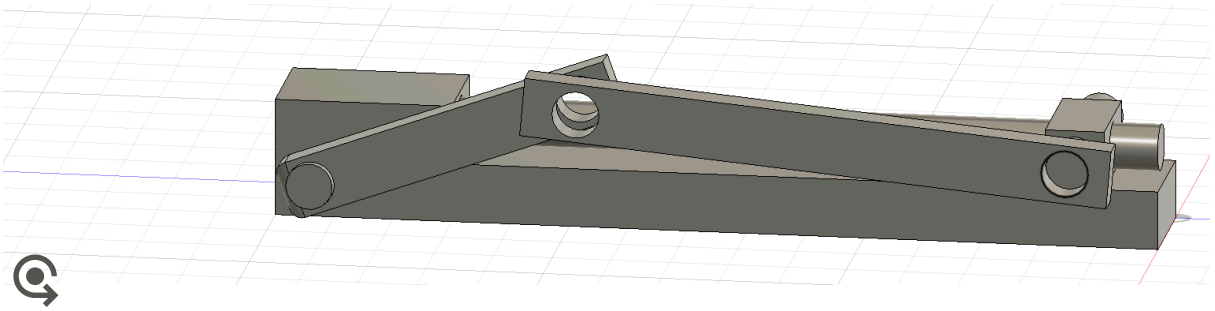

Mechanically transfer circular movement to a linear movement.

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Date	2025 autumn
School	Malmö University

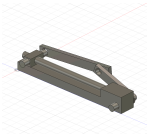

Figure 1. Hero Picture.

Overview

This project explores the mechanical translation of circular motion into controlled linear movement using a 3D-printed rod-and-lever system. The design went through multiple iterations to resolve friction and rotation lock-ups, including redesigning the lever joints and introducing custom-printed threaded parts and fasteners. Printed in PLA for fast iteration, the final prototype runs continuously for over 10 minutes without failure and clearly demonstrates linear motion in both still images and testing documentation.

Reflection

This really pushed me the most out of all the projects so far. I had to redesign the same parts over and over again, and the first two full attempts simply didn't work. I didn't understand the mechanic at all in the beginning. But the more I tested, watched examples, asked for help and actually paid attention to what was happening, the more things clicked.

By the end, I finally understood how to control the circular-to-linear motion and how to avoid blocking my own mechanism. It became easier once the logic sank in, and honestly this ended up being the project I learned the most from. Even though it's a quick and dirty prototype, it did exactly what it needed to do, and building it taught me more than any of the smoother, easier assignments.

Visuals

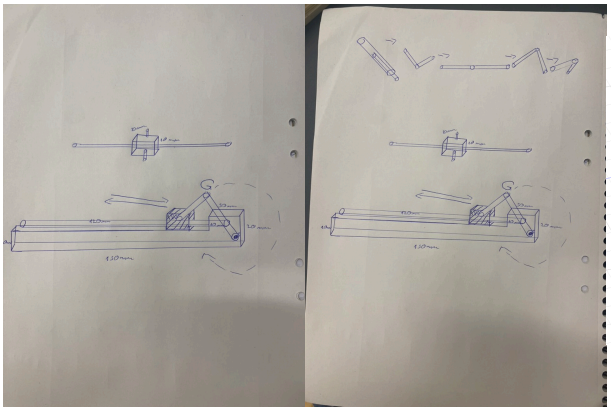

Figure 2. Sketches.

Figure 3. Sketching rod movement.

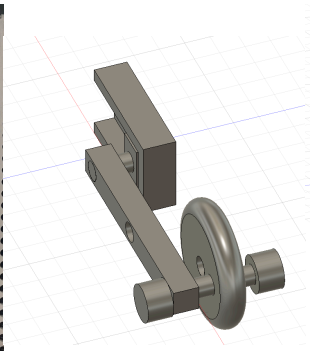

Figure 4. First try shape.

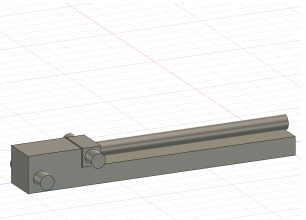

Figure 5. Squares, connections.

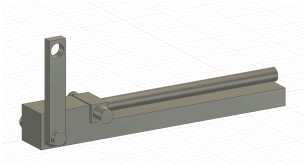

Figure 6. The fundamental square, rod, moving square.

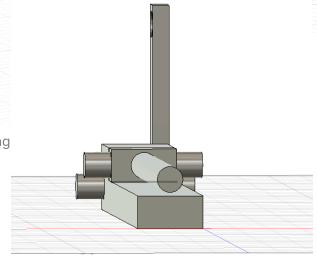

Figure 7. (Figure 6)- from different view.

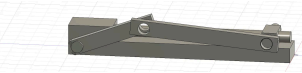

Figure 8. Both rod, showing the movement workflow.

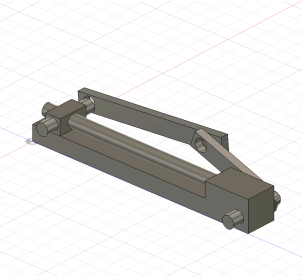

Figure 9. (Figure 8)- from other viewpoint.

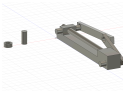

Figure 10. The cylinder to connect the rods and holder.

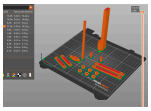

Figure 11. Preparation for printing

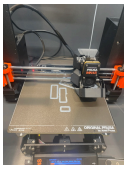

Figure 12. Printing supporting material.

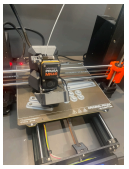

Figure 13. Printing.

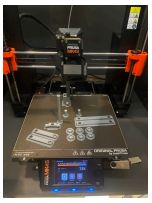

Figure 14. Printing is almost done.

Figure 15. (Figure 14) - from other perspective.

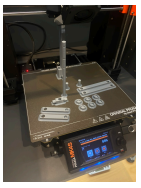

Figure 16. Done print.

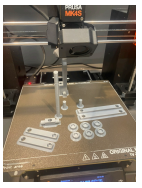

Figure 17. (Figure 16) - from other view.

Figure 18. First printed items test pieces.

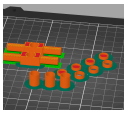

Figure 19. Starting fixing the issues with rotation and other minor errors.

Figure 20. Lego with printed female and normal srews.

Figure 21. Product with screws.

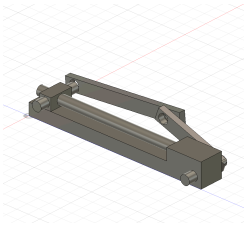

Figure 22. Final product.

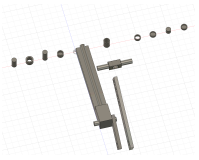

Figure 23. All items needed to make it work.

Figure 24. Working prototype. Linear motion visible.

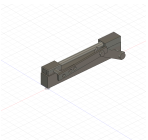

Figure 25. Made it again, with only one side rod for more clear view.

Figure 26. Using one side in the rode to avoid errors with two rods.